

Task W05: Doubly-Linked Lists

Data Structures, Grade Sheet

Name:

Date:

Language used (circle one): Java C/C++

Submission and setup

3 pts

Submitted the entire labw05 project (Eclipse project for Java, full folder including the Makefile for C/C++), not just individual files

/ 1

Project compiles and runs without errors

/ 1

Driver's existing file-reading code left in place; own testing code added below the marked point

/ 1

Code quality

3 pts

Naming and formatting are reasonable and consistent with the starter code

/ 1

No leftover dead code, stray debugging output, or commented-out blocks

/ 1

No unrelated changes to Node, the interface, or the exception class

/ 1

Part 1: Finish the Conversion

18 pts

add() keeps prev and tail correct through head, middle, and end insertions

/ 2

remove(0) sets the new head's prev to null / nullptr

/ 2

remove() on a non-head node fixes the successor's prev

/ 2

A new list's tail starts null / nullptr

/ 2

tail is set after adding the first item to an empty list

/ 2

tail is updated when add() appends after the last node

/ 2

tail is updated when remove() removes the last node

/ 2

tail becomes null / nullptr when remove() empties the list

/ 2

removeAll() clears both head and tail

/ 2

Part 2: New Methods

12 pts

displayBackward() prints the list in reverse

/ 2

addFirst() adds to the front and keeps prev/tail correct

/ 2

addLast() adds to the end and keeps prev/tail correct

/ 2

addLast() does not call add() internally (read the code; not caught by the test script)

/ 2

removeFirst() removes the front item, keeps prev/tail correct, and throws on an empty list

/ 2

removeLast() removes the back item, keeps prev/tail correct, and throws on an empty list

/ 2

Task W05: Doubly-Linked Lists (continued)

Part 3: Put Your List to Work

18 pts

isPalindrome() correct on an empty list		/ 2
isPalindrome() correct on a single-item list		/ 2
isPalindrome() correct on an even-length palindrome		/ 2
isPalindrome() correct on an odd-length palindrome		/ 2
isPalindrome() correct on a non-palindrome		/ 2
reverse() correct on an empty list		/ 2
reverse() correct on a single-item list		/ 2
reverse() correct on an even-length list, prev/tail intact		/ 2
reverse() correct on an odd-length list, prev/tail intact		/ 2

Total

/ 54

Notes